AC9M6A02 • Year 6 Mathematics

Unknown Values with Brackets and Mixed Operations

Use equality, operation properties and inverse reasoning

We are learning to…

Students interpret equality as balance, follow grouping, identify the role of the unknown and use inverse operations or number properties before substituting to check.

Represent | Reason | Calculate | Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

find unknown values in numerical equations involving brackets and combinations of arithmetic operations, using the properties of numbers and operations

Solve 4(□ + 7) = 100

1

Interpret

4 equal groups total 100

2

Undo ×4

□ + 7 = 25

3

Undo +7

□ = 18

4

Substitute

4(18 + 7) = 100

5

Confirm

both sides equal

Undo operations in reverse order while respecting brackets. The method follows the structure of the expression rather than moving symbols without meaning.

Use properties to simplify before solving

6(□ + 4) = 180

divide by 6, then subtract 4

3□ + 5□ = 96

combine like multiplicative groups: 8□ = 96

5(20 − □) = 35

divide, then solve 20 − □ = 7

(□ + 8) ÷ 4 = 9

multiply by 4, then subtract 8

The unknown may appear in a subtracted position, so operation order and meaning matter. Always check in the original equation.

Build and annotate the model

Represent unknown values with brackets and mixed operations and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Operations undone in written order:** Undo the outer operation first.
- **Bracket ignored:** Treat the grouped expression as one quantity.
- **Subtraction assumed commutative:** $20 - x$ is not $x - 20$.
- **Check performed on a simplified but altered equation:** Substitute into the original.

Quick check

- Solve $4(x+7)=100$.
- Solve $5(20-x)=35$.
- Combine $3x+5x$.
- Explain reverse order.
- Substitute and check.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.